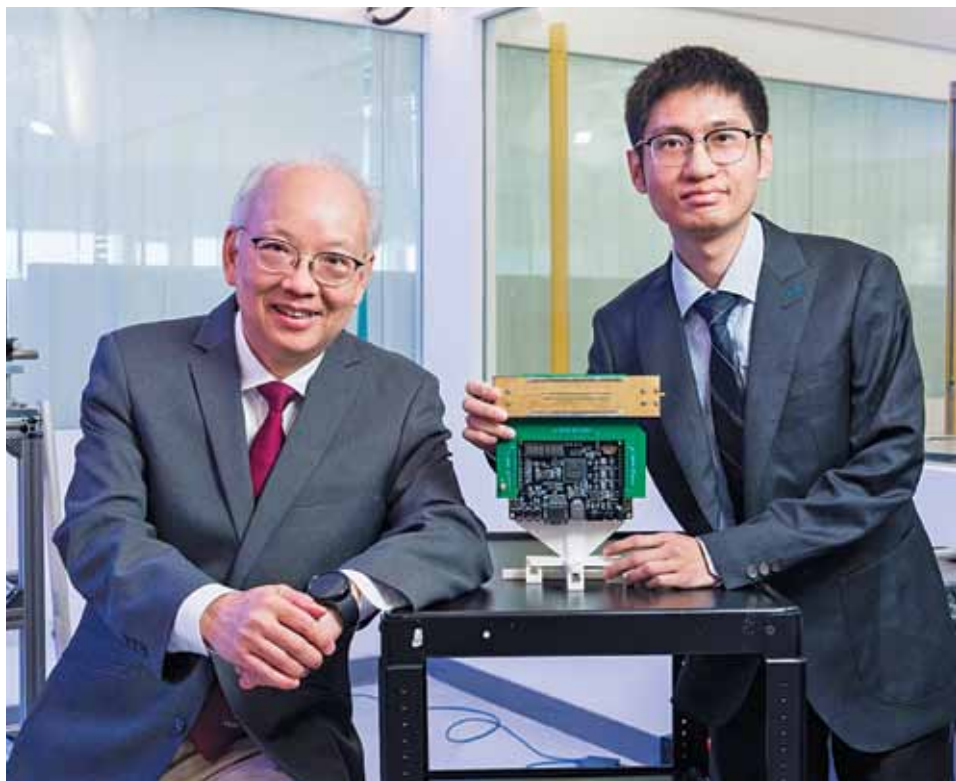




RESEARCH & INNOVATION UPDATES

Universal metasurface antenna for high security 6G communications

The research team at the City University of Hong Kong has achieved a breakthrough in antenna technology. They created a metasurface antenna capable of controlling all fundamental properties of electromagnetic waves through software, marking a historic milestone in electromagnetic engineering. This achievement enables the manipulation of electromagnetic radiation without adjectives like amplitude, phase, frequency, polarisation, and direction. As global research progresses towards 6G wireless communication, this antenna offers promise for diverse applications. Its waveform manipulation and security features are crucial for integrating sensing and communications, revolutionising information systems, real-time imaging, and wireless power transfer. The antenna's direction modulation enhances privacy and security. This innovation is set to shape the future of wireless communication and information systems in the 6G era.



Prof. Chan Chi-hou (left) and Prof. Wu Gengbo (right) showcase the universal metasurface antenna developed at CityU, which allows unprecedented manipulation of electromagnetic waves (Credit: City University of Hong Kong)

Budget-friendly silicon solar cells achieve 26.4% efficiency

Researchers at Suzhou Maxwell Technologies Co. Ltd, Soochow, New South Wales, and Dalian University have presented an approach to silicon heterojunction (SHJ) solar cell fabrication, achieving 26.4% power conversion efficiency. SHJ solar cells offer high efficiency, cost-effective processing, and low-temperature coefficients but face scalability and cost challenges. The team tackled these challenges by replacing traditional fabrication methods. They substituted conventional doped a-Si:H with doped hydrogenated nanocrystalline silicon (nc-Si:H) and its

alloys with oxygen (nc-SiOx:H) and carbon (nc-SiC:H) to reduce parasitic absorption and series resistance. They also utilised very high-frequency (VHF) plasma-enhanced chemical vapour deposition (PECVD) systems for faster deposition and seed-free plated-copper electrodes, improving current density. These innovations promise scalable, affordable, and efficient SHJ solar cells to combat carbon emissions and climate change. High-efficiency SHJ solar cells are a significant step toward sustainable energy and a greener future.